

Lab: Brain Plasticity -- How Your Brain Rewires Itself

Objective

Learning Goal: Investigate neuroplasticity by testing how your brain changes with practice, exploring how sleep affects learning, and discovering how the nervous system physically rewires when you learn.

This is an **investigation lab** designed for middle school students (Grades 6-8).

Materials Needed

- Paper and pencil
 - Mirror (for mirror drawing activity)
 - Timer or stopwatch
 - Lab journal or notebook
-

Part 1: Mirror Drawing Challenge (15 min)

When you learn something new, your brain physically creates new neural connections. Let's watch this happen in real time by trying to draw while looking only in a mirror.

Setup: Place a mirror in front of you. Try to trace a simple star shape while looking **ONLY** at your reflection in the mirror (not at your hand directly). Do this 5 times.

Attempt	Time to complete (seconds)	Errors (times you went outside the lines)	Frustration level (1-10)
1			
2			
3			
4			
5			

Write your response here

Part 2: Neuroplasticity Case Studies (10 min)

Your brain changes its physical structure when you learn. This is called **neuroplasticity**. Discuss these real cases with your group.

Case	What happened in the brain?	Active Inference connection
London taxi drivers have larger hippocampi from memorizing street maps		
Musicians have more gray matter in areas controlling their instrument hand		
People who lose their sight develop enhanced hearing		
Learning a second language changes brain structure		

Write your response here

Part 3: Sleep and Learning Investigation (10 min)

Sleep is when your brain consolidates learning -- strengthening new neural connections and pruning weak ones. Survey your group about sleep and learning.

Question	Your Answer	Group Average
How many hours of sleep did you get last night?		
Did you study anything yesterday?		
How well do you remember it today (1-10)?		
Do you ever have better ideas after sleeping on a problem?		

Write your response here

Part 4: Reflection Table

Question	Your Answer
What is neuroplasticity?	
How does your brain physically change when you learn?	
Why did mirror drawing get easier with practice?	
How does sleep help learning?	
How does Active Inference explain brain plasticity?	

Write your response here